MPG No. 21.3.78 6′ [Type III, hexagonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & T_{xxy} & 0 \\ -T_{xxx} & -T_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxy} & -T_{xxx} & 0 \end{bmatrix}$$

$T_{xxx} = \mathbb{T}^{(1)}_{3c}$
$T_{xxy} = \mathbb{T}^{(1)}_{3s}$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & 0 & 0 & -T_{xxzx} \\ T_{xxzx} & -T_{xxzx} & 0 & 0 & 0 & T_{xxyz} \\ 0 & 0 & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{T}^{(1)}_{4s}$
$T_{xxzx} = \mathbb{T}^{(1)}_{4c}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & T_{xxzx} \\ -T_{xxzx} & T_{xxzx} & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = 2\mathbb{M}^{(1)}_{3c}$
$T_{xxzx} = -2\mathbb{M}^{(1)}_{3s}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & T_{xxzx} & 0 \\ -T_{xxyz} & -T_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{xxzx} & -T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{M}^{(2)}_{3c}$
$T_{xxzx} = \mathbb{M}^{(2)}_{3s}$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} \\ T_{yzxy} & -T_{yzxy} & 0 & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$T_{yzxx} = \mathbb{M}^{(3)}_{3c}$
$T_{yzxy} = \mathbb{M}^{(3)}_{3s}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & T_{xxzx} & 0 & T_{xxzy} & T_{xxxz} & 0 \\ 0 & 0 & 0 & -T_{xxyz} & -T_{xxzx} & 0 & -T_{xxzy} & -T_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\ -T_{yzxy} & T_{yzxy} & 0 & 0 & 0 & T_{yzxx} & 0 & 0 & T_{yzxx} \\ 0 & 0 & 0 & -T_{xxxz} & T_{xxzy} & 0 & -T_{xxzx} & T_{xxyz} & 0 \end{bmatrix}$$

$$\begin{aligned} T_{xxyz} &= 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\ T_{xxzx} &= -2\mathbb{M}_{3s}^{(1)} + \mathbb{M}_{3s}^{(2)} + \mathbb{T}_{4c}^{(1)} \\ T_{xxzy} &= 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)} \\ T_{xxxz} &= -2\mathbb{M}_{3s}^{(1)} - \mathbb{M}_{3s}^{(2)} + \mathbb{T}_{4c}^{(1)} \\ T_{yzxx} &= -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)} \\ T_{yzxy} &= -2\mathbb{M}_{3s}^{(1)} - \mathbb{T}_{4c}^{(1)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzxy} \\ T_{yzxy} & -T_{yzxy} & 0 & 0 & 0 & -T_{yzxx} & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{yzxx} &= \mathbb{M}_{3c}^{(3)} \\ T_{yzxy} &= \mathbb{M}_{3s}^{(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & T_{xxxz} \\ 0 & 0 & T_{yyyz} \\ 0 & 0 & 0 \\ -T_{xxxz} & T_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -T_{yyyz} \\ 0 & 0 & 0 \\ T_{xxxz} & -T_{yyyz} & 0 \\ 0 & 0 & -T_{xxxz} \\ -T_{yyyz} & -T_{xxxz} & 0 \end{bmatrix}$$

$$\begin{aligned} T_{xxxz} &= \mathbb{T}_{4c}^{(1)} \\ T_{yyyz} &= -\mathbb{T}_{4s}^{(1)} \end{aligned}$$